STATISTICS

PART 8: Confidence Interval (for two populations)

Birol Aslanyürek

Rev: 12.11.2024

CONFIDENCE INTERVALS FOR TWO POPULATIONS

- The population parameters and corresponding estimators for two populations are given below:

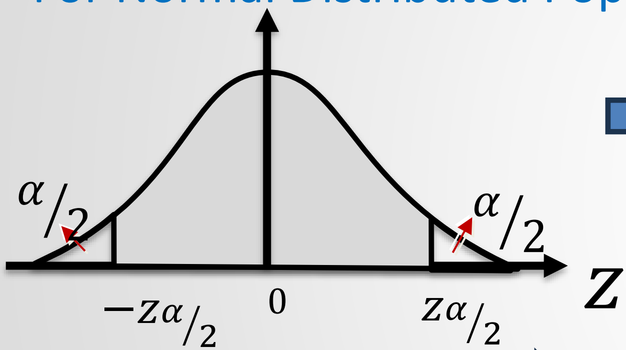
Population Parameter	**Estimator $\hat{\theta}$**	**Estimate value $\hat{\theta}$**
Difference between Means $(\mu_1 - \mu_2)$	$\bar{X}_1 - \bar{X}_2$	$\bar{x}_1 - \bar{x}_2$
Difference between Means for Paired Populations (μ_D)	$\bar{D}$	$\bar{d}$
Ratio of variances $\left({\sigma_1^2}/{\sigma_2^2}\right)$	${S_1^2}/{S_2^2}$	${s_1^2}/{s_2^2}$
Difference between proportions $(P_1 - P_2)$	$\hat{P}_1 - \hat{P}_2$	$\hat{p}_1 - \hat{p}_2$

Confidence Interval

Statistics

The Confidence Interval for the Difference between Population Means

For Normal Distributed Populations with Known Variances



$$\Rightarrow \mathrm{P}\left(-z_{\alpha/2} \le \frac{(\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}} \le z_{\alpha/2}\right) = 1-\alpha$$

To extract $\mu_1-\mu_2$ in inequality, …

$$\Rightarrow \mathrm{P}\left(-z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}} \le (\bar{X}_1-\bar{X}_2)-(\mu_1-\mu_2) \le z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}\right) = 1-\alpha$$

$$\Rightarrow \mathrm{P}\left(-(\bar{X}_1-\bar{X}_2)-z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}} \le \mu_1-\mu_2 \le -(\bar{X}_1-\bar{X}_2)+z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}\right) = 1-\alpha$$

$100(1-\ \alpha)\%$ confidence interval for $\mu_1-\mu_2$

$$\Rightarrow \left((\bar{X}_1-\bar{X}_2)-\underbrace{z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}_{e},(\bar{X}_1-\bar{X}_2)+\underbrace{z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}_{e}\right)$$

Confidence Interval

Statistics

for normal distributed populations with unknown variance $\sigma_1^2 = \sigma_2^2 = \sigma^2$

$$P\left(-t_{\alpha/2,n-1} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_k\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \leq t_{\alpha/2,n-1}\right) = 1 - \alpha$$

$$\left[(\bar{X}_1 - \bar{X}_2) - \underbrace{t_{\frac{\alpha}{2},n_1+n_2-2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}_{e}, (\bar{X}_1 - \bar{X}_2) + \underbrace{t_{\frac{\alpha}{2},n_1+n_2-2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}_{e}\right]$$

for normal distributed populations with unknown variance $\sigma_1^2 \neq \sigma_2^2$

$$P\left(-t_{\frac{\alpha}{2},v} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \leq t_{\frac{\alpha}{2},v}\right) = 1 - \alpha$$

$$\left[(\bar{X}_1 - \bar{X}_2) - \underbrace{t_{\frac{\alpha}{2},v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}_{e}, (\bar{X}_1 - \bar{X}_2) + \underbrace{t_{\frac{\alpha}{2},v} \sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}_{e}\right]$$

Confidence Interval

Statistics

for unknown distributed populations with known variances ($n_1 \geq 30$ and $n_2 \geq 30$)

$$P\left(-z_{\alpha/2} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$\left[(\bar{X}_1 - \bar{X}_2) - \underbrace{z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}_{e}, (\bar{X}_1 - \bar{X}_2) + \underbrace{z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}_{e}\right]$$

for unknown distributed populations with unknown variances ($n_1 \geq 30$ and $n_2 \geq 30$)

$$P\left(-z_{\alpha/2} \leq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

$$\left[(\bar{X}_1 - \bar{X}_2) - \underbrace{z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}_{e}, (\bar{X}_1 - \bar{X}_2) + \underbrace{z_{\alpha/2}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}_{e}\right]$$

Confidence Interval

Statistics

Example 1: To compare the nicotine content of two cigarette brands, 10 cigarettes are selected from the first brand and 8 cigarettes are selected from the second brand. The mean and standard deviation of the first sample are 3.1 mg and 0.5 mg, respectively, while the mean and standard deviation of the second sample are 2.7 mg and 0.7 mg, respectively. By assuming that both samples are selected from the populations distributed normally with equal variances, construct a 95% confidence interval for the difference between the two brand means.

$n_1 = 10 \quad n_2 = 8 \quad S_1 = 0.5 \quad S_2 = 0.7 \quad \bar{X}_1 = 3.1 \quad \bar{X}_2 = 2.7$

$X_1 \sim N, X_2 \sim N \quad \sigma_1^2 = \sigma_2^2 \text{ (unknown)} \qquad t_{\frac{\alpha}{2}, n_1+n_2-2} = t_{0.025,16} = 2.120$

$$S_k = \sqrt{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}} = \sqrt{\frac{(10-1)0.5^2 + (8-1)0.7^2}{10+8-2}} = 0.596$$

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2}, n_1+n_2-2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2}, n_1+n_2-2} S_k \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}\right] = [-0.2, 1.0]$$

Thus, we can claim that the difference between the two brands' means is between the interval [-0.2, 1.0] with a confidence of 95%. Also, there is no significant difference between means since the confidence interval contains zero.

Confidence Interval

Statistics

Example 2: A study was conducted by the Department of Zoology at Virginia Tech to estimate the difference in the amounts of the chemical orthophosphorus measured at two different stations on the James River. Orthophosphorus was measured in milligrams per liter. Fifteen samples were collected from station 1, and 12 samples were obtained from station 2. The 15 samples from station 1 had an average orthophosphorus content of 3.84 milligrams per liter and a standard deviation of 3.07 milligrams per liter, while the 12 samples from station 2 had an average content of 1.49 milligrams per liter and a standard deviation of 0.80 milligrams per liter. Find a 95% confidence interval for the difference in the mean of orthophosphorus contents at these two stations, assuming that the observations came from normal populations with different variances.

$n_1 = 15$

$\bar{X}_1 = 3.84$

$S_1 = 3.07$

$n_2 = 12$

$\bar{X}_2 = 1.49$

$S_2 = 0.80$

$\alpha = 1 - 0.95 = 0.05$

$X_1 \sim N, X_2 \sim N$

$\sigma_1^2 \neq \sigma_2^2$ (unknown)

Confidence Interval

Statistics

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}} = \frac{\left(\frac{3.07^2}{15} + \frac{0.80^2}{12}\right)^2}{\frac{\left(\frac{3.07^2}{15}\right)^2}{15 - 1} + \frac{\left(\frac{0.80^2}{12}\right)^2}{12 - 1}} = 16.3 \approx 16$$

$$\alpha = 0.05, \qquad t_{\frac{\alpha}{2},\nu} = t_{0.025,16} = 2.120$$

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2},\nu}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2},\nu}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}\right]$$

$$= \left[(3.84 - 1.49) - 2.12\sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}}, (3.84 - 1.49) + 2.12\sqrt{\frac{3.07^2}{15} + \frac{0.80^2}{12}}\right]$$

$$0.60 \leq \mu_1 - \mu_2 \leq 4.10$$

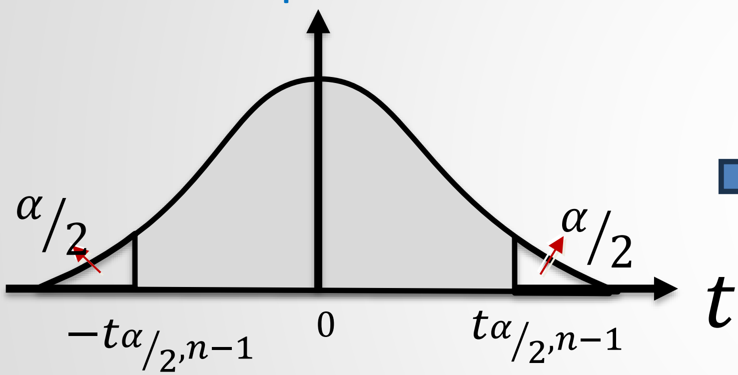
The difference between the two brands’ means is between the interval [0.60, 4.10] with a confidence of 95%. There is a significant difference between means since the confidence interval does not contain zero.

Confidence Interval

Statistics

The Confidence Interval for the Difference between Population Means

For Paired Populations Distributed Normally…



$$\Rightarrow \mathrm{P}\left(-t_{\alpha/2,n-1} \le \frac{\bar{D}-\mu_D}{S_D/\sqrt{n}} \le t_{\alpha/2,n-1}\right) = 1-\alpha$$

To extract μ_D in inequality, …

$$\Rightarrow \mathrm{P}\left(-t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}} \le \bar{D}-\mu \le t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}}\right) = 1-\alpha$$

$$\Rightarrow \mathrm{P}\left(-\bar{D}-t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}} \le -\mu \le -\bar{D}+t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}}\right) = 1-\alpha$$

$$\Rightarrow \mathrm{P}\left(\bar{D}-t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}} \le \mu \le \bar{D}+t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}}\right) = 1-\alpha$$

$100(1-\alpha)\%$ confidence interval for μ_D

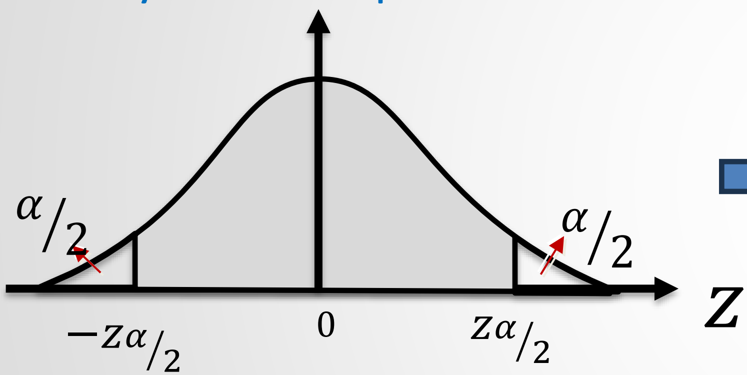
$$\Rightarrow \left(\bar{D}-t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}},\ \bar{D}+t_{\alpha/2,n-1}\,{S_D}/{\sqrt{n}}\right)$$

Confidence Interval

Statistics

The Confidence Interval for the Difference between Population Means

For Any Paired Populations with $n \geq 30$



$$\Rightarrow P\left(-z_{\alpha/2} \leq \frac{\bar{D} - \mu_D}{S_D/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

To extract μ_D in inequality, …

$$\Rightarrow P\left(-z_{\alpha/2}\, S_D/\sqrt{n} \leq \bar{D} - \mu \leq z_{\alpha/2}\, S_D/\sqrt{n}\right) = 1 - \alpha$$

$$\Rightarrow P\left(-\bar{D} - z_{\alpha/2}\, S_D/\sqrt{n} \leq -\mu \leq -\bar{D} + z_{\alpha/2}\, S_D/\sqrt{n}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\bar{D} - z_{\alpha/2}\, S_D/\sqrt{n} \leq \mu \leq \bar{D} + z_{\alpha/2}\, S_D/\sqrt{n}\right) = 1 - \alpha$$

$100(1 - \alpha)\%$ confidence interval for μ_D

$$\Rightarrow \left(\bar{D} - z_{\alpha/2}\, S_D/\sqrt{n}, \bar{D} + z_{\alpha/2}\, S_D/\sqrt{n}\right)$$

Confidence Interval

Statistics

Example 3: To test the effect of an online tutorial on the students, 10 students are randomly chosen and their grades are measured before and after the online tutorial.

Students	After the online tutorial	Before the online tutorial
Student_1	62	61
Student_2	71	76
Student_3	66	70
Student_4	87	91
Student_5	69	46
Student_6	77	80
Student_7	45	37
Student_8	61	73
Student_9	89	90
Student_10	65	70

Find the 95% confidence interval for the difference between the mean of grades before and after the online tutorial. (Assume that populations have normal distributions.)

Confidence Interval

Statistics

If populations were independent, …

$$n_1 = n_2 = 10$$

$$\overline{X_2} = \frac{\sum_{i=1}^{n} X_{2i}}{n_2} = 69.4$$

$$\overline{X_1} = \frac{\sum_{i=1}^{n} X_{1i}}{n_1} = 69.2$$

$$S_1^2 = \frac{\sum_{i=1}^{n}(X_{1i} - \overline{X_1})^2}{n_1 - 1} \cong 167.29$$

$$S_2^2 = \frac{\sum_{i=1}^{n}(X_{2i} - \overline{X_2})^2}{n_2 - 1} \cong 303.16$$

for unequal variances,…

$$\nu = \frac{\left(\frac{S_1^2}{n_1}+\frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1-1}+\frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2-1}} \cong 16.61 \cong 17$$

$$t_\nu = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

$$t_{\frac{0.05}{2},17} = 2.11$$

$$\left[(\bar{X}_1 - \bar{X}_2) - t_{\frac{\alpha}{2},\nu}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}, (\bar{X}_1 - \bar{X}_2) + t_{\frac{\alpha}{2},\nu}\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}\right] = [-14.67, 14.27]$$

INCORRECT SOLUTION !,
SINCE
POPULATIONS ARE DEPENDENT

Confidence Interval

Statistics

Students	X_{1i}	X_{2i}	$D_i = X_{1i} - X_{2i}$
Student_1	62	61	1
Student_2	71	76	-5
Student_3	66	70	-4
Student_4	87	91	-4
Student_5	69	46	23
Student_6	77	80	-3
Student_7	45	37	8
Student_8	61	73	-12
Student_9	89	90	-1
Student_10	65	70	-5

$$\bar{D} = \frac{\sum_{i=1}^{n} D_i}{n} = -0.2$$

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \bar{D})^2}{n-1}} \cong 9.60$$

$$t_{n-1} = \frac{\bar{D} - \mu_D}{S_D / \sqrt{n}} \qquad t_{10-1,0.05/2} = 2.26$$

$$\left[\bar{D} - t_{n-1,\alpha/2}\frac{S_D}{\sqrt{n}}, \bar{D} + t_{n-1,\alpha/2}\frac{S_D}{\sqrt{n}}\right]$$

$$= [-7.09, 6.67]$$

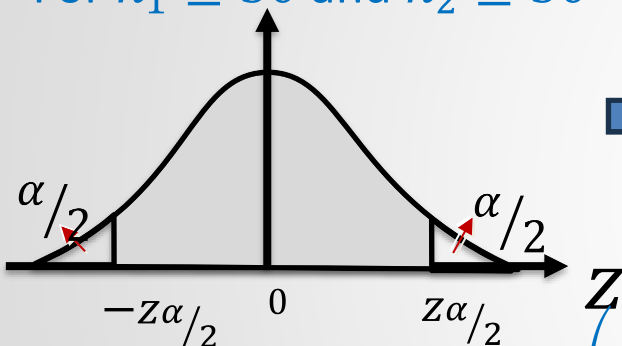
CORRECT SOLUTION

Confidence Interval

Statistics

The Confidence Interval for the Difference between Population Proportions

For $n_1 \geq 30$ and $n_2 \geq 30$



$$\Rightarrow \quad P\left(-z_{\alpha/2} \leq \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

To extract $p_1 - p_2$ in inequality, …

$$\Rightarrow \quad P\left(-z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} \leq (\hat{p}_1 - \hat{p}_2) - (p_1 - p_2) \leq z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}\right) = 1 - \alpha$$

$$\Rightarrow \quad P\left(-(\hat{p}_1 - \hat{p}_2) - z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} \leq p_1 - p_2 \leq -(\hat{p}_1 - \hat{p}_2) + z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}\right) = 1 - \alpha$$

$100(1 - \alpha)\%$ confidence interval for $p_1 - p_2$

$$\Rightarrow \quad \left((\hat{p}_1 - \hat{p}_2) - \underbrace{z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}_{e}, (\hat{p}_1 - \hat{p}_2) + \underbrace{z_{\alpha/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}}_{e}\right)$$

Confidence Interval

Statistics

Example 4: 132 of the 200 male voters and 90 of the 150 female voters support a specific candidate. Find a 99% confidence interval for the difference in the exact proportion of male and female voters that support that candidate.

$n_1 = 200$

$n_2 = 150$

$X_1 = 132$

$X_2 = 90$

$$\hat{p}_1 = \frac{X_1}{n_1} = \frac{132}{200} = 0.66$$

$$\hat{p}_2 = \frac{X_2}{n_2} = \frac{90}{150} = 0.60$$

$$\hat{q}_1 = \frac{68}{200} = 0.34$$

$$\hat{q}_1 = \frac{60}{150} = 0.40$$

$$z\alpha_{/2} = z_{0.005} = 2.575$$

$$\left((\hat{p}_1 - \hat{p}_2) - z\alpha_{/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}, (\hat{p}_1 - \hat{p}_2) + z\alpha_{/2}\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}} \right)$$

$$-0.074 \leq p_1 - p_2 \leq 0.194$$

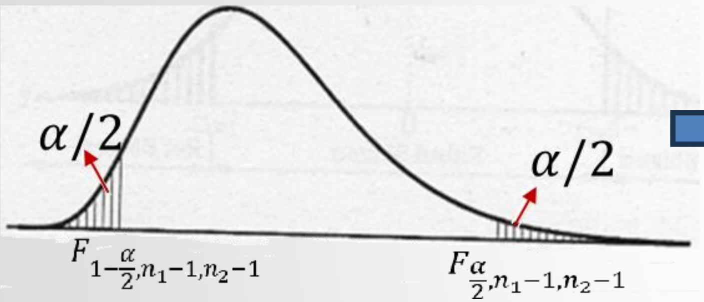
The difference between the exact proportion of male and female voters that support that candidate is between the interval [-0.074, 0.194] with a confidence of 99%.

Confidence Interval

Statistics

The Confidence Interval for the Ratio of Two Population Variances

For Normal Distributed Populations



$$P\left(F_{1-\frac{\alpha}{2},n_1-1,n_2-1} \le \frac{S_1^2\sigma_2^2}{S_2^2\sigma_1^2} \le F_{\frac{\alpha}{2},n_1-1,n_2-1}\right) = 1-\alpha$$

To extract ${\sigma_1^2}/{\sigma_2^2}$ in inequality, …

$$P\left(\frac{1}{F_{\frac{\alpha}{2},n_1-1,n_2-1}} \le \frac{S_2^2\sigma_1^2}{S_1^2\sigma_2^2} \le \frac{1}{F_{1-\frac{\alpha}{2},n_1-1,n_2-1}}\right) = 1-\alpha$$

$$F_{1-\frac{\alpha}{2},n_1-1,n_2-1} = \frac{1}{F_{\frac{\alpha}{2},n_2-1,n_1-1}}$$

$$P\left(\frac{1}{F_{\frac{\alpha}{2},n_1-1,n_2-1}}\frac{S_1^2}{S_2^2} \le \frac{\sigma_1^2}{\sigma_2^2} \le F_{\frac{\alpha}{2},n_2-1,n_1-1}\frac{S_1^2}{S_2^2}\right) = 1-\alpha$$

$100(1-\ \alpha)\%$ confidence interval for ${\sigma_1^2}/{\sigma_2^2}$

$$\left[\frac{1}{F_{\frac{\alpha}{2},n_1-1,n_2-1}}{S_1^2}/{S_2^2},\ F_{\frac{\alpha}{2},n_2-1,n_1-1}{S_1^2}/{S_2^2}\right]$$

Confidence Interval

Statistics

Example 5: To compare the nicotine content of two cigarette brands, 10 cigarettes are selected from the first brand and 8 cigarettes are selected from the second brand. The mean and standard deviation of the first sample are 3.1 mg and 0.5 mg, respectively, while the mean and standard deviation of the second sample is 2.7 mg and 0.7 mg, respectively. Construct a 98% confidence interval for the proportion of the two brands' exact variances.

$n_1 = 10$ $\quad n_2 = 8$ $\quad X_1 \sim N, X_2 \sim N$ $\quad F_{\frac{\alpha}{2}, n_1-1, n_2-1} = F_{0.01,9,7} = 6.72$

$\bar{X}_1 = 3.1$ $\quad \bar{X}_2 = 2.7$ $\quad \alpha = 1 - 0.98 = 0.02$ $\quad F_{\frac{\alpha}{2}, n_2-1, n_1-1} = F_{0.01,7,9} = 5.61$

$S_1 = 0.5$ $\quad S_2 = 0.7$

$$\left[\frac{1}{F_{\frac{\alpha}{2}, n_1-1, n_2-1}} {S_1^2}/{S_2^2} \, , F_{\frac{\alpha}{2}, n_2-1, n_1-1} {S_1^2}/{S_2^2}\right] \Rightarrow \frac{1}{6.72}\frac{0.5^2}{0.7^2} \le \frac{\sigma_1^2}{\sigma_2^2} \le 5.61\frac{0.5^2}{0.7^2}$$

$$\Rightarrow 0.076 \le \frac{\sigma_1^2}{\sigma_2^2} \le 2.862$$

The ratio of two brands' exact variances is between the interval [0.076, 2.862] with a confidence of 98%.